

Soil structural interfaces in some Texas Vertisols and their impact on solute transport

M. Morgun Nobles*, L.P. Wilding, K.J. McInnes

Department of Soil and Crop Sciences, Texas A&M University, College Station, TX, USA

Abstract

Structural surfaces and slickensides have been reported to be important features impacting sorption–desorption of chemicals and bypass flow of pollutants to shallow groundwater aquifers. This study provides visible, microscopic and submicroscopic characteristics of slickenside features in two Texas Vertisols of different geomorphological age. Transport of visible anionic dye and CaBr is studied as related to soil structural surfaces, root channels and slickenside planes. Macromorphic observations were coupled with thin section microscopy, scanning electron microscopy and electron microprobe analyses.

Slickenside surfaces differed from subjacent micromass of the ped interiors by stronger preferred orientation of clay particles and reduction in macropores with corresponding increase in micropores. Such characteristics could make slickensides act as semi-permeable barriers to solute transport, but this phenomenon could not be confirmed from CaBr studies herein.

The majority of the solute bypass flow in both soils was conducted through root channels and ped interfaces. Slickenside surfaces in these Vertisols conducted less than 1% of the flow. It was observed primarily when the dye solution was delivered to the surface of the slickenside via the adjacent root channel. In spite of the preferential orientation of clay particles along slickenside planes, the penetration of Br into the aggregate interior was not retarded like the larger organic anion dye. These results may reflect specific moisture conditions, which should be considered in future works.

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Keywords: Vertisol; Slickenside; Preferential flow; Structural surfaces

1. Introduction

The presence of slickensides in well-structured soils has been observed visually and microscopically, but few studies of submicroscopic properties of these features have

* Corresponding author. Fax: +1-979-862-1712.

E-mail address: m-morgun@tamu.edu (M. Morgun Nobles).

been made (Sullivan, 1994). The formation of slickenside features is thought to be caused by shrinking and swelling of confined soil material rich in clay, resulting in shear failure within the soil fabric. This phenomenon leads to a reorientation and denser packing of soil plasma, creating lineated zones with face to face clay alignment (Wilding and Tessier, 1988). These separations were generally characterized by vosepic, skelsepic, and possibly lattisepic plasmic fabrics (Mermut et al., 1996). Blokhuis et al. (1990) noted that the surface-related plasma separations in Vertisols reached maximum development in the zone of maximum slickenside concentrations.

Slickenside planes were shown to be one of the primary pathways of solute movement in the soil (Lin, 1995). When slickensides are common throughout subsoil material, the sum of their microfabric effects can influence macroscopic fluxes of solutes. It was hypothesized that during a flow event, slickenside surfaces would act as a physical barrier against solute movement into and out of the soil matrix because of the smaller pore size and parallel or subparallel reorientation of clay particles. This hypothesis is supported by findings of Lai and Mortland (1968), who established that the coefficient for Na diffusion flux perpendicular to the clay platelets was only 15% to 20% of the diffusion coefficient for flux parallel to clay platelets. It is possible that the effect would increase proportionally to the thickness of the oriented layer.

This study investigates the micromorphological properties of structural surfaces such as root channels and slickenside planes of two Vertisols that might influence the movement of solutes in these soils. The study also explores the relationship between structural and micromorphological properties of soil and the transport of solutes at macroscopic (0.1 m²), microscopic (100 cm²) and submicroscopic (1 mm²) scales. Conducting the research on these particular soils provided us with the opportunity to study Vertisols that contained a range in soil age and structural properties, particularly slickenside features.

2. Materials and methods

The study was conducted on two soils with vertic properties (Ships and Burleson series) located in the Blackland Land Resource Area in Central Texas (Fig. 1). The moisture and temperature regimes of the area are classified as ustic and thermic (Mowery et al., 1958). The Ships series was located on a nearly level floodplain of the Brazos River formed several hundred years ago. Previous research showed that macropore flow existed along some slickenside surfaces in these soils (Lin, 1995). It contained abundant, but not continuous, slickensides. It was characterized as fine, mixed, superactive, thermic Chromic Haplusterts. The site of Burleson series was located on the terrace of Brazos River formed approximately 30,000 years ago. It had abundant large slickenside planes and normal gilgai relief. Some slickenside planes achieved the dimensions of 80 cm to 1 m. It was characterized as fine, smectitic, thermic Udic Haplusterts. Selected properties of these soils are presented in Table 1.

In order to examine the pathways of solute transport in these soils at macroscopic, microscopic and submicroscopic scales, they were treated with solution of Brilliant

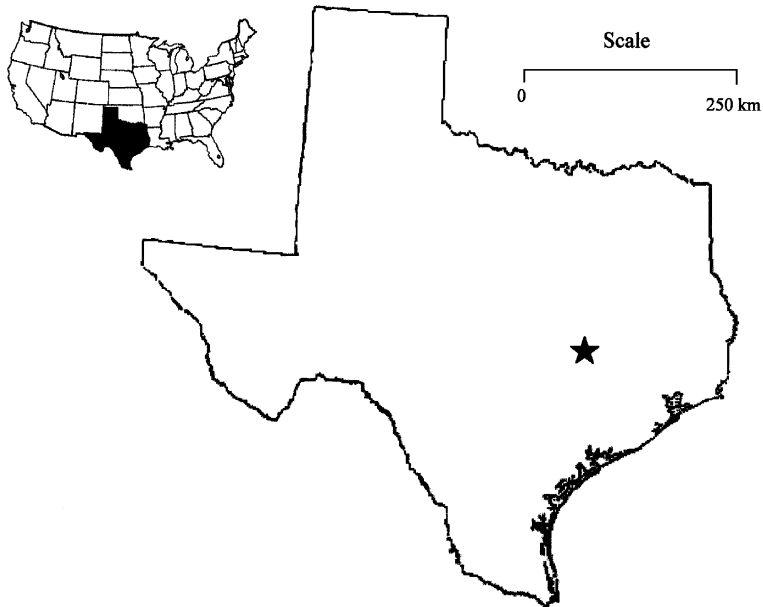


Fig. 1. Location of soils selected for this study.

Blue FCF dye and CaBr. Brilliant Blue FCF dye is a water-soluble anionic organic dye of dark blue color, highly visible in most soils (Flury and Flühler, 1995). Due to the larger molecule size, Brilliant Blue FCF could serve as an indicator of pesticide

Table 1
The soils investigated in this study and their basic properties

Horizon	Depth, cm	Texture	Total clay, %	CEC, cmol/kg	Organic carbon, %	CEC/clay, cmol/kg	pH (1:1 H ₂ O)	CCE ^a %
<i>Ships clay (fine, mixed, superactive, thermic Chromic Haplusterts)</i>								
Ap1	0–10	Silty Clay Loam	32.4	26.0	1.09	80	8.0	12.7
Ap2	10–25	Silty Clay Loam	37.7	28.2	0.96	75	8.0	13.7
Bssk1	25–60	Silty Clay Loam	38.8	27.2	0.67	70	8.0	12.2
Bssk2	60–80	Silty Clay	41.4	29.6	0.70	71	8.0	11.8
Bssk3	80–120	Clay	61.8	41.1	0.86	67	7.9	10.3
<i>Burleson clay (fine, smectitic, thermic Udic Haplusterts)^b</i>								
Microhigh								
A	0–20	Clay	50.6	45.0	1.42	89	7.6	11.3
Bssk1	20–63	Clay	53.9	43.0	0.45	80	7.8	12.9
Bssk2	63–147	Clay	52.1	42.5	0.14	82	7.9	18.9
Microlow								
A	0–20	Clay	60.9	57.5	1.98	94	7.3	0.4
BA	20–50	Clay	63.2	51.9	1.30	82	7.4	1.1
Bssk	50–166	Clay	61.3	48.7	0.95	79	7.6	3.2

^a CCE—Calcium Carbonate Equivalent.

^b Lin (1995).

and herbicide movement in soil, while Br is often used as an anionic tracer to indicate the movement of water in soil, as well as potential movement of nitrates.

Four PVC cylinders with an inner diameter of 35 cm were inserted in soil to a depth of 15 cm. Fifty-three liters of a tracing solution of Brilliant Blue FCF at a concentration of 30 g l^{-1} (Flury and Flühler, 1995) and CaBr at a concentration of 60 g l^{-1} were applied to each of the three cylinders. The same quantity of aqueous solution containing only Brilliant Blue FCF at a concentration of 30 g l^{-1} was applied to the fourth cylinder. It was later used to obtain the control samples for antecedent Br concentration identification. Fifty-three liters constituted approximately one pore volume of a soil column 35 cm in diameter to the depth of 1.5 m, based on a dry soil. Dye solution was manually applied to the cylinders with the head of approximately 5 cm. The values of soil moisture tension of Ships clay and Burleson clay at the time of solute application varied from 1000 to 1500 kPa. The absence of open desiccation cracks in Both Ships clay and Burleson clay was visually observed at the time of the experiment.

The soil was excavated through the center of the dye application zone within several hours after tracing solution completely infiltrated into soil, and the exposed flow paths were examined. Dye patterns were traced onto transparent plastic sheets (Petersen et al., 1997; Lin, 1995) and color photos of dye patterns were taken. Morphological features involved in the transport of dye, such as ped interfaces, biopores and slickenside planes, were identified in the field and color-labeled on plastic sheets.

The tracings were later transferred to drafting velum and scanned into a computer. The color images were then converted to black and white binary images and analyzed. The total area of dye-stained section was calculated and the percentage of dye-stained area contributed by each type of feature was estimated as a fraction of total area for every soil horizon and for the entire profile.

Samples of natural soil aggregates were collected from subsurface horizons with particular attention to structural features such as slickensides and biopores. The orientation of samples was recorded for later reference. The diameter of samples ranged from 5 to 10 cm. The samples were stored in the freezer until they were freeze-dried in Virtis 25 EL freeze-dryer. Thin sections were prepared to a $30 \mu\text{m}$ thickness after impregnation with a 10:4 mixture of Shell EPON epoxy resin and EpiCure 3274 curing agent (Innes and Pluth, 1970).

Thin section slides were examined with a petrographic microscope to observe the orientation and the thickness of the slickensides (Soil Survey Staff, 1996). Thin section slides were described according to the terminology developed by Brewer (1964) and Bullock et al. (1985). Stress features in thin sections of Ships clay and Burleson clay were partially obscured by the presence of pedogenic carbonates. To allow for the observation of previously obliterated features, several thin sections were decalcified by submersion of the sample in a beaker containing cold 1 N HCl until all carbonates were removed from the section (Wilding and Drees, 1990).

Fractured blocks containing both planar and cross-sectional orientation of slickenside surfaces were coated with Au–Pd and studied with a JEOL T330 scanning electron microscope (Goldstein et al., 1992). Particular attention was paid to the

thickness and density of the layer of oriented clay particles and their relative degree of orientation.

Polished blocks containing the structural features stained by the dye were attached to small petrographic slides (27×46 mm) and cut into thin sections. Resulting samples were coated with carbon by means of evaporation (Reed, 1996) and studied with Cameca SX50 Electron Microprobe (Cameca FRANCE, Courbevoie, France) using wavelength-dispersive capabilities (WDS) to obtain Br distribution across the surface of the sample, as related to the surfaces of the structural features such as slickenside surfaces and root channels. Voltage for WDS analysis was set at 15 kV, while the beam current used was 40 nA. The diameter and depth of electron beam contact were about 1 μm .

Linear traverses were conducted on 20 samples containing root channels, 12 samples containing root interfaces and 5 samples containing slickenside planes. The distribution in the number of subcategories reflects the relative number of dye tracings observations associated with these morphological features. The resulting count rate for Br was plotted against the traverse distance from a given structural reference feature. The general trend in Br distribution was represented by a sixth order polynomial curve. The curves of Br distribution for selected samples were compared with the optical image of the thin section.

3. Results and discussion

3.1. Micromorphological properties of Ships clay

The matrix of the Bssk horizon of Ships clay was characterized by the presence of pedogenic carbonates, such as common calcans and many irregular carbonate nodules set in a porphyroclastic groundmass (open porphyric related distribution) and areas of crystic plasmic fabric (crystallic b-fabric). After decalcification of selected thin sections, the dominance of vo-masepic (porostriated b-fabric) plasmic fabric in Ships clay became more evident. The size of microaggregates was estimated as 0.3–0.5 mm, bounded by stress-oriented features (granostriated b-fabric) 5–10 μm in thickness.

The presence of stress-oriented clay features was particularly evident in the vicinity of slickenside surfaces (Fig. 2) and at ped interfaces. The stress-oriented structures were identified as thin (5 μm) bands of plasmic material located parallel or subparallel to the slickenside in the close vicinity of the slickenside surface. The distribution of stress-oriented features became much less ordered in the ped interior.

Compound meta-voids, channels, chambers, skew planes and vughs were predominant in Ships clay. The matrix included common normal and irregular manganiferous nodules with sharp boundaries. Isolated islands of skeleton grains occurred throughout the matrix in the lower part of the Bssk horizon.

The SEM analysis showed that slickenside surfaces of Ships clay had micro-grooves approximately 0.3 to 20 μm wide (Fig. 3). The grooves were thought to be caused by physical movement of silt-sized skeleton grains being translated parallel to the direction of shear failure. At the slickenside surfaces, clay particles were oriented parallel or

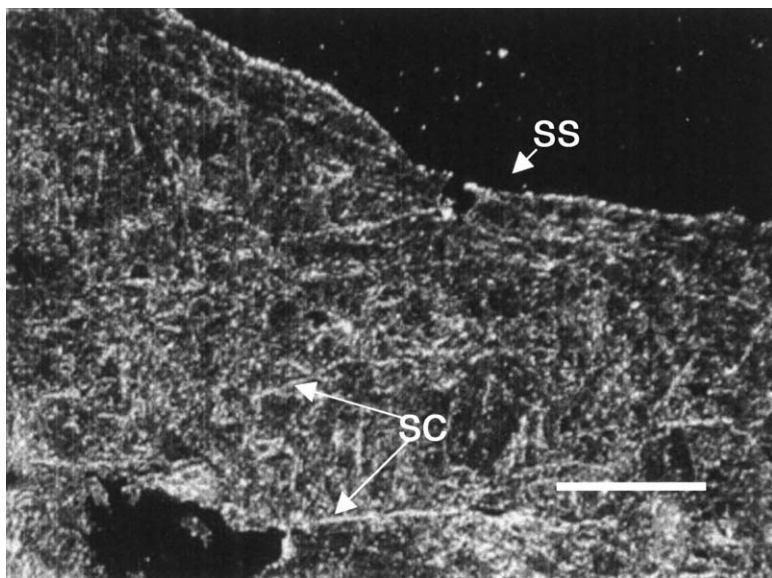


Fig. 2. Microphotograph from decalcified thin sections of the Bssk horizon of Ships clay showing stress-induced plasma reorientation adjacent to the surface of the slickenside (SS) and plasma preferred orientation (SC) in intrapedal matrix. Cross-polarized light. Bar length is 500 μm .

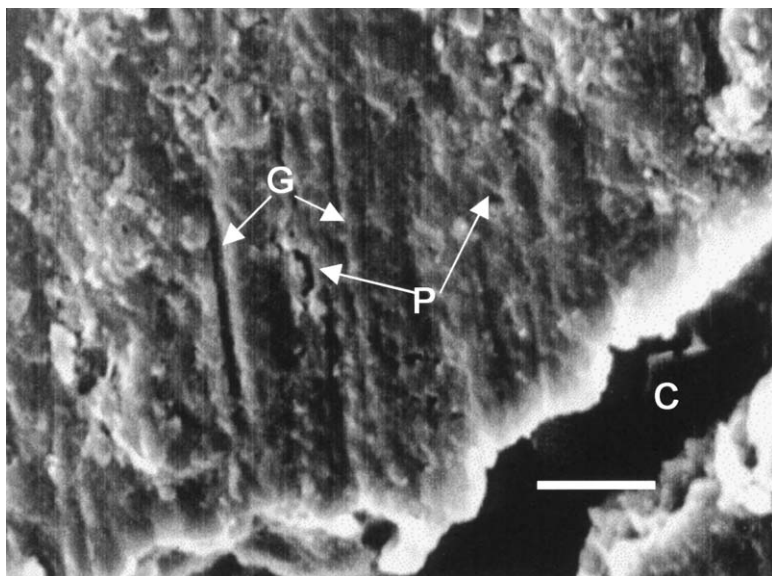


Fig. 3. Scanning electron microphotograph of slickenside surface (planar view) in the Bssk horizon of Ships clay showing grooving (G) and coarse crack (C). Bar length is 5 μm .

subparallel to the surface (Fig. 4). The thickness of oriented clay was only about 3 to 4 μm , but continuous over distances of at least 500 μm (the width of the SEM slide). Under the field conditions reoriented clay layer might appear to be continuous for distances up to 1 m.

It was noted by Lai and Mortland (1968) that the diffusion coefficient for Na flux perpendicular to clay platelets was only 15% to 20% of the diffusion coefficient for flux parallel to clay platelets. It is possible that the strong degree of clay particle orientation at the surfaces of slickensides in Vertisols is capable of producing the similar effects on solute movement. However, it should be recognized that the results observed by Lai and Mortland (1968) were obtained during a laboratory scale experiment with simulated platelets without consideration for desiccation. In contrast, slickenside planes are less perfectly oriented and are subject to wetting and drying.

The presence of an oriented clay layer had a direct impact on the orientation and size distribution of pores and cracks. Over a 1200 μm^2 area of a scanned slickenside surface of Ships clay (planar view), slickensides had pores that were generally less than 5 μm in diameter and were intersected by coarse cracks about 10 μm in width (Fig. 3). The subjacent matrix in Ships clay was characterized by frequent pores ranging in size from 5 to 25 μm and channels ranging from 3 to 10 μm (Fig. 4). The presence of pores and cracks on the generally smooth slickenside surfaces suggests that these features may be used as conduits for solute transport across the otherwise poorly penetrable clay layer, thus directly influencing the transport of solutes into and out of ped interior.

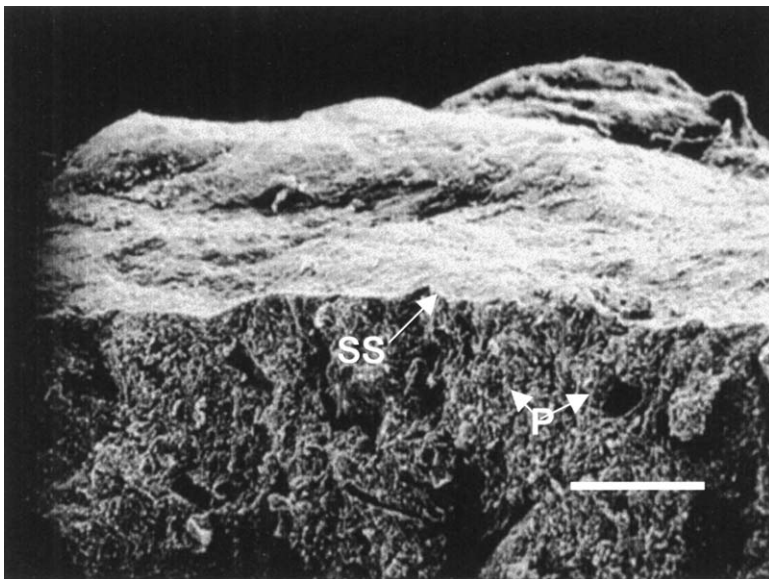


Fig. 4. Scanning electron microphotograph of a fractured soil block from the Bssk horizon in Ships clay, showing a cross-section view of plasma preferred orientation at the slickenside surface (SS) and pores (P) in the subjacent matrix. Bar length is 100 μm .

3.2. Micromorphological properties of Burleson clay

Pedogenic carbonates in form of carbonate nodules were also detected in the Bssk horizon of Burleson clay. However, unlike the plasmic fabric (micromass) of Ships clay, the concentration of fine-grained carbonates in Burleson clay was not sufficient to form a crystic plasmic fabric. This allowed for the direct observation of micromorphological features in the thin sections from the Bssk horizon of Burleson clay without decalcification of thin sections. The plasmic fabric of Burleson clay was found to be skel-vo-masepic (grano-poro-striated b-fabric). The s-matrix (groundmass) of Burleson clay was porphyro-skelic (open porphyric related distribution). Plasma preferred orientations were readily observed throughout the matrix of Burleson clay as thin (5–10 μm) bands of plasmic material. The presence of these features was particularly evident in the vicinity of the slickenside surfaces and ped interfaces where stress cutans were oriented parallel or subparallel along the aggregate surface (Fig. 5). Stress-oriented features were also detected along void and channel walls and surfaces of skeleton grains in the form of granostriated and porostriated b-fabric.

The voids present in Burleson clay were compound meta-voids, channels, chambers and vughs. The presence of meta-craze planes was also detected. The s-matrix (groundmass) of Burleson clay included common normal and irregular manganiferous nodules of various sizes. In general, Fe–Mn nodules had rounded shapes and abrupt boundaries. Few Fe–Mn nodules were surrounded by simple stress cutans (granostriated b-fabric).

The surfaces of slickenside planes in the Burleson clay had microgrooves approximately 1 to 20 μm wide (Fig. 6). The width of some of the micro-grooves on

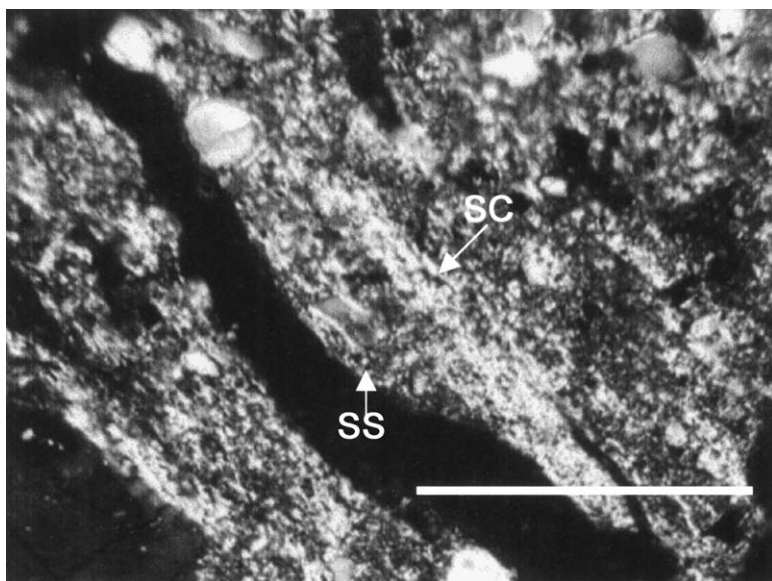


Fig. 5. Micrograph from a thin section of the Bssk horizon of Burleson clay showing a simple stress cutan (SC) located subparallel to the ped surface. Cross-polarized light. Bar length is 100 μm .

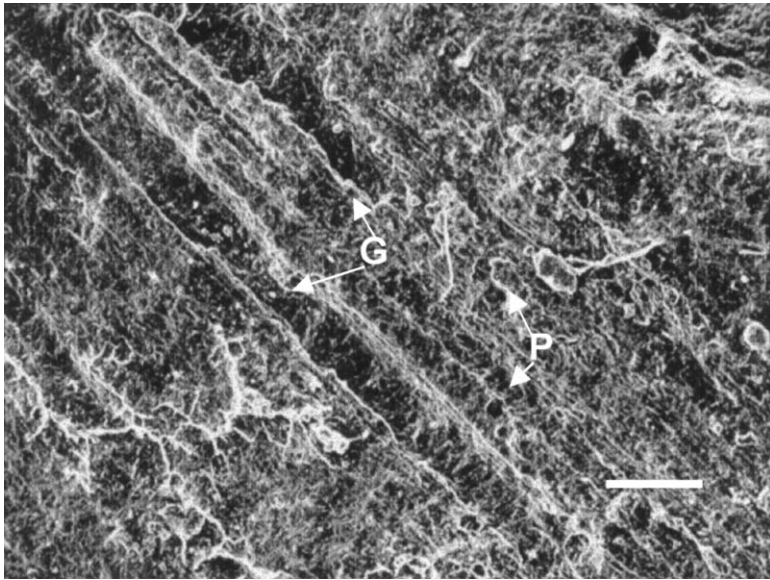


Fig. 6. Scanning electron micrograph of a planar view of slickenside surface in the Bssk horizon of Burleson clay, showing grooving (G). Bar length is 100 μm .

slickenside surfaces reached 50 μm . Similar to the slickenside surfaces of Ships clay, clay particles at the slickenside surfaces of Burleson clay were found to be oriented parallel or subparallel to slickenside surfaces (Fig. 7). The thickness of the oriented clay layer was comparable to that in Ships clay and did not exceed 5 μm . However, it was observed that the slickenside surfaces of Burleson clay were penetrated by pores of larger diameter, such as 50 μm , with some pores reaching a diameter of 200 μm . The underlying matrix was characterized by pores and channels ranging in size from 4 to 50 μm . The surfaces of the slickenside in Burleson clay were intersected by coarse cracks about 10 μm in width, with some cracks wider than 100 μm . It is possible that some of these coarse cracks, particularly the cracks wider than 50 μm , were artificially widened during the slow freezing procedure.

3.3. Micromorphological observations of roots and root channels

The presence of many roots in fine interaggregate pores in Ships clay was evident from thin section observations, as well as the SEM analysis. The diameter of root cross-sections varied from 0.2 to 0.5 mm (Fig. 8). Root cross-sections and root channels were not as prominent in the thin sections of Burleson clay as in Ships clay. However, a few roots were detected in thin sections of these soils. The diameter of root cross-sections in thin sections of Burleson clay varied from 0.5 to 1 mm.

In the process of thin section preparation, the roots seemed to have shrunk and pulled away from the root channel wall to some degree, thereby rendering it difficult to describe the initial position of roots as related to the walls of pore channels.

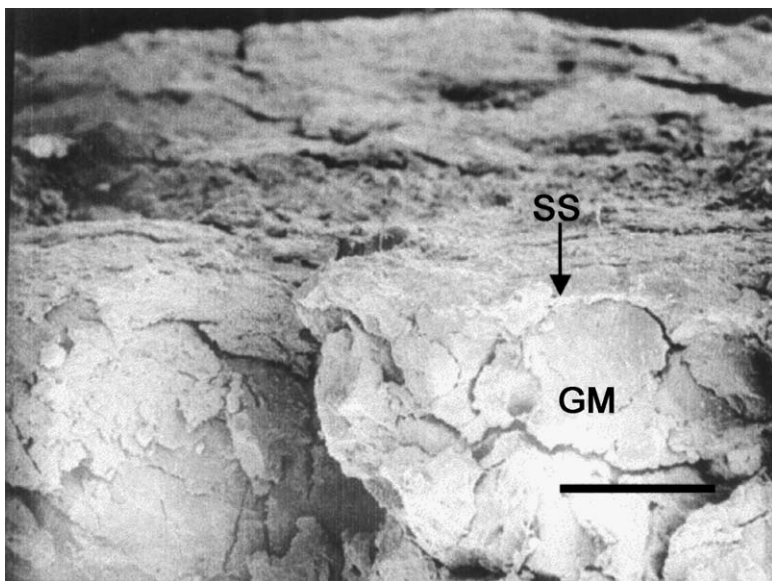


Fig. 7. Scanning electron micrograph of the fractured soil aggregate from the Bssk horizon of Burleson clay (normal view) showing the cross-section view of the slickenside surface (SS). Lower part of micrograph shows the view of the groundmass (GM). Bar length is 500 μm .

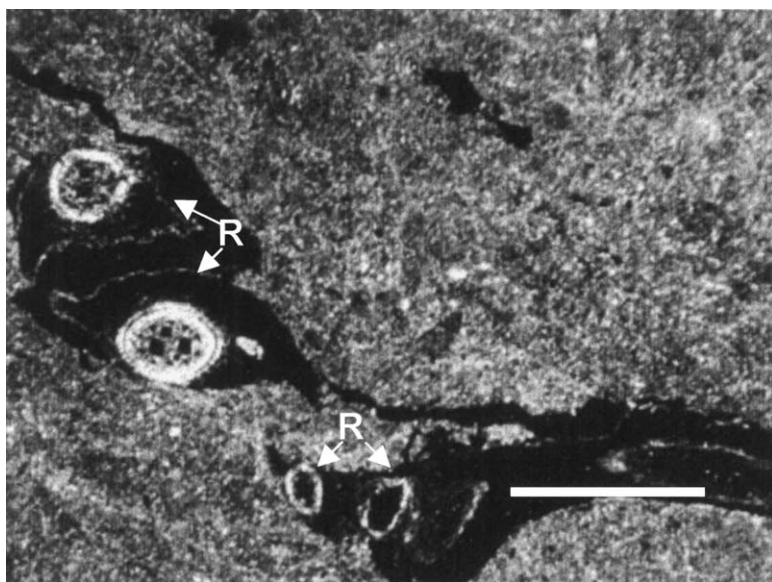


Fig. 8. Thin section micrograph showing the position of roots (R) in root channels of Bssk horizon (60 cm) of Ships clay. Cross-polarized light. Bar length is 500 μm .

However, from visible observations the general position of roots in root channels suggested that the root-associated pores were not completely occupied by living roots, leaving a void between the root and some parts of the wall of root channel. This observation is supported by findings of Kooistra et al. (1992), which showed that the average fraction of root–soil contact varied from 60% to 87% depending on the degree of soil compaction. The presence of the partial void between the root and the wall of the root channel would allow root channels to serve as readily available pathways for soil solutes.

Presence of roots in all soils was also evident from the SEM observations. Many roots, some flattened, were positioned between abutting slickenside fissures (Fig. 9A). This was also a common field observation. This suggested that the stress-oriented clay layer at the surface of the slickenside could serve as a barrier for the root penetration through the slickenside plane into the ped matrix. In several cases, however, roots were found penetrating the surface of the slickenside, forming root channels approximately 200 to 250 μm wide into the soil matrix (Fig. 9B).

3.4. Morphological observations of dye movement in soils

In subsurface horizons of four Ships clay sites, stained areas were mainly associated with ped interfaces (approximately 18% of total stained area) and root channels (approximately 37% of total stained area) (Fig. 10). Less than 1 cm^2 (<1% of total stained area) of the stained surfaces in the subsoil horizons in one site of Ships clay was attributed to the flow along the slickenside planes. However, the flow along the slickenside faces was visually observed in the other three sites of Ships clay during collection of the samples. The quantification of the dye stain was limited to the center of the dye application zone and was not conducted for the collected samples.

Involvement of slickenside surfaces in the solute transport was primarily detected when the dye solution was transported to the surface of slickenside by the adjacent root channels located in the immediate vicinity of the slickenside fissure. Commonly, the solution moved along very fine cylindrical root channels penetrating the surface of the slickenside. In such cases, the contact of the root channel with the slickenside fissure would cause the dye solution to advance along the surface of the slickenside and stain a small (approximately 3–4 mm) area on the surface of the slickenside immediately adjacent to the root channel (Fig. 11). The primary transport of the solution would then continue along the root channel. In some instances, stained root channels were positioned between abutting slickenside surfaces (Fig. 11). The spatial distribution of dye on the surface of slickenside in these cases was often limited to the vicinity of the root channel, without the movement of the dye solution further along the slickenside surface.

In subsurface horizons of one site of Burleson clay microhigh and two sites of Burleson clay microlow, stained areas were mainly associated with ped interfaces (approximately 50% in Burleson clay microlows) and root channels (58% in Burleson clay microhigh). Slickenside surfaces contributed to approximately 6 cm^2 (1% of total stained area) of stained area in the Bsk1 horizon and 2 cm^2 (8%) in the Bsk2 horizon of Burleson clay microhigh. In addition to dye-stained slickenside faces detected during

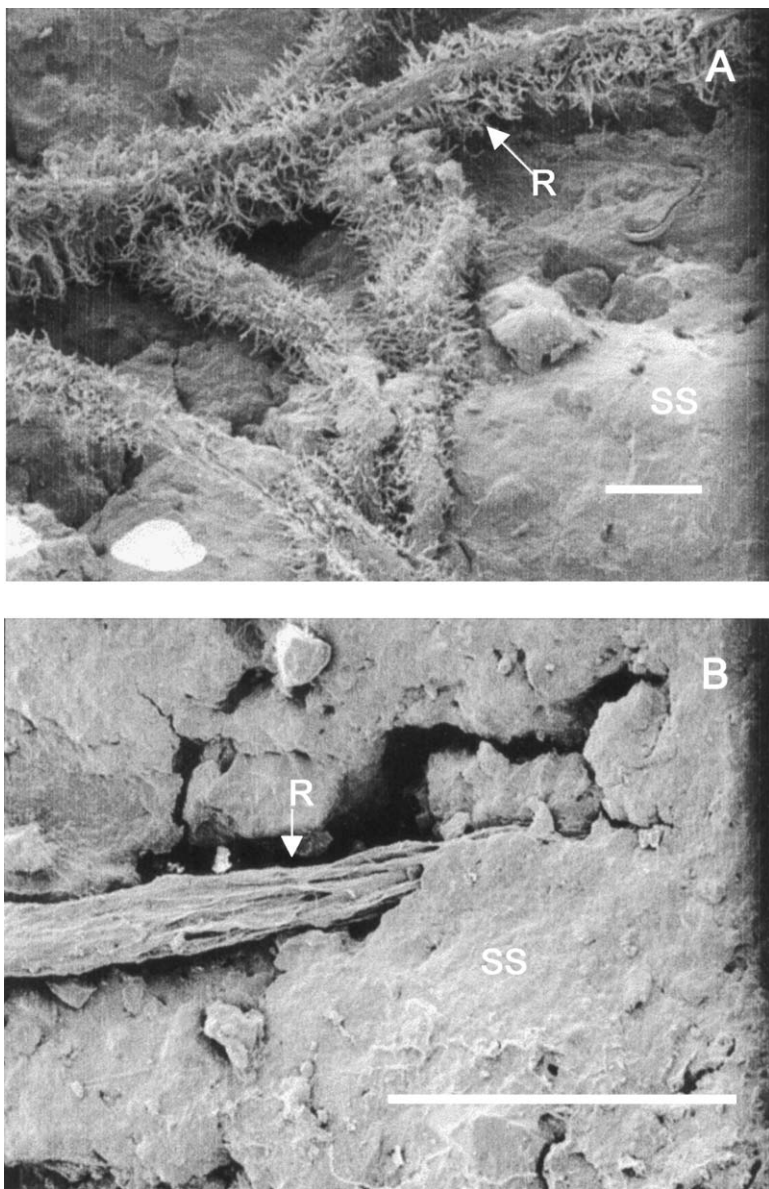


Fig. 9. Scanning electron microphotographs showing the position of roots (R) on planar slickenside surfaces (SS) in Ships clay (A) and Burleson clay (B). Bar length is 500 μm .

tracing of the dye pattern, slickenside planes stained with dye were observed later during sampling. Solute transport along these features was conducted almost exclusively when the dye solution was delivered to the surface of the slickenside through the adjacent root channel. The spatial distribution of dye in these cases did not appear

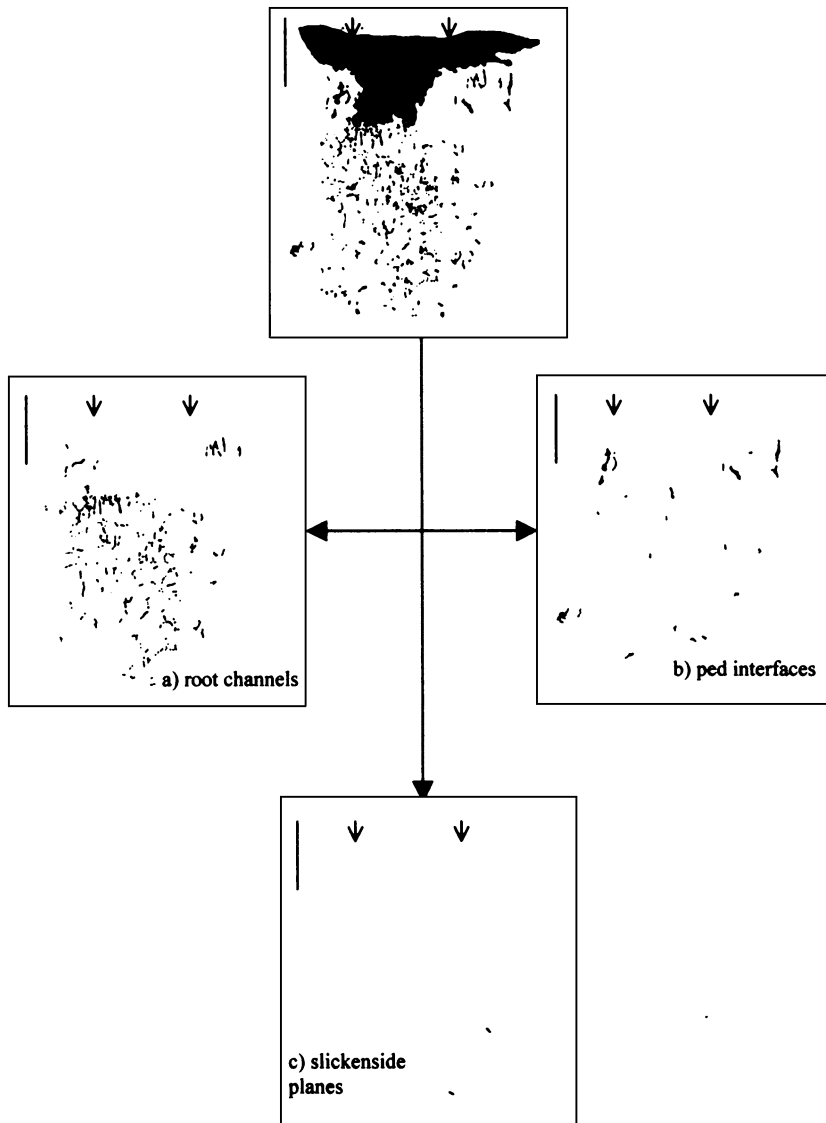


Fig. 10. Two-dimensional dye flow pattern in site 4 of Ships clay exposed along the central vertical line of intake zone (indicated by arrows). The vertical bar indicates 20 cm.

to be limited to the confines of the root channel, with the dye solution moving along the slickenside surface. This was true when the root was positioned between slickenside planes, as well as when very fine roots penetrated the surfaces of the slickensides.

Less than 1 cm^2 of stained area ($< 1\%$ of total stained area) in the Bssk horizon of site 2 of Burleson clay microlow was associated with solute flow along slickenside surfaces.

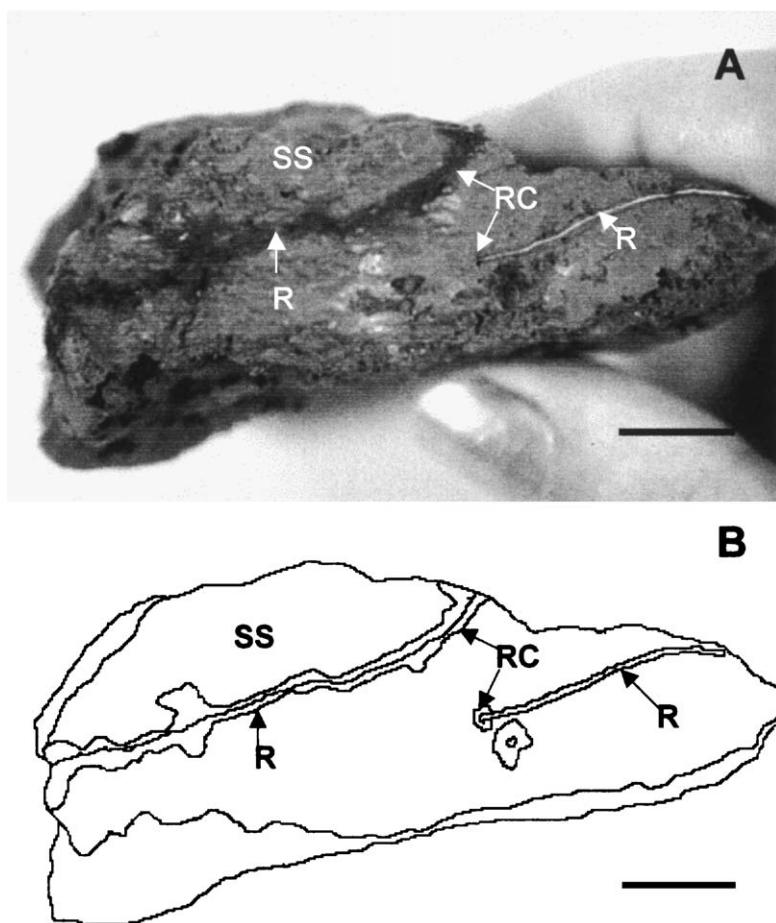


Fig. 11. (A) Slickenside plane (SS) in the Bssk horizon of Ships clay stained by Brilliant Blue FCF dye that has been transported to the slickenside surface by root channels (RC). Position of roots is indicated by (R). (B) Line drawing of the soil clod in (A). Bar length is 1 cm.

However, this was the only case of solute flow along the slickenside surface observed in Burleson clay microlow. There were no other cases of flow along slickensides detected in the remaining portion of site 1 or in the entire site 2.

It is possible that the lack of open desiccation cracks in these Vertisols have resulted in the decreased participation of slickenside fissures in the solute transport at the time of this study. The absence of open desiccation cracks has direct influence on the type of solute pathways. Kutilek (1996) noted that the cracks are the predominant channels available for deep moistening of Vertisols. When cracking clay soils are wetted, the majority of cracks are transformed into micropores of similar character as the interpedal pores, which could cause the increase in the involvement of root channels and interpedal pores. The absence of open cracks in both Ships clay and Burleson clay was visually

observed at the time of the experiment. It is likely that an increase in involvement of desiccation cracks and slickenside planes in solute transport may occur under drier conditions. The involvement of slickenside planes could also become more prominent in cases of high intensity, short duration rainfall events. Extra solute volume could create depressions partially filled with water and will result in higher hydrostatic heads, which in turn will increase slickenside involvement.

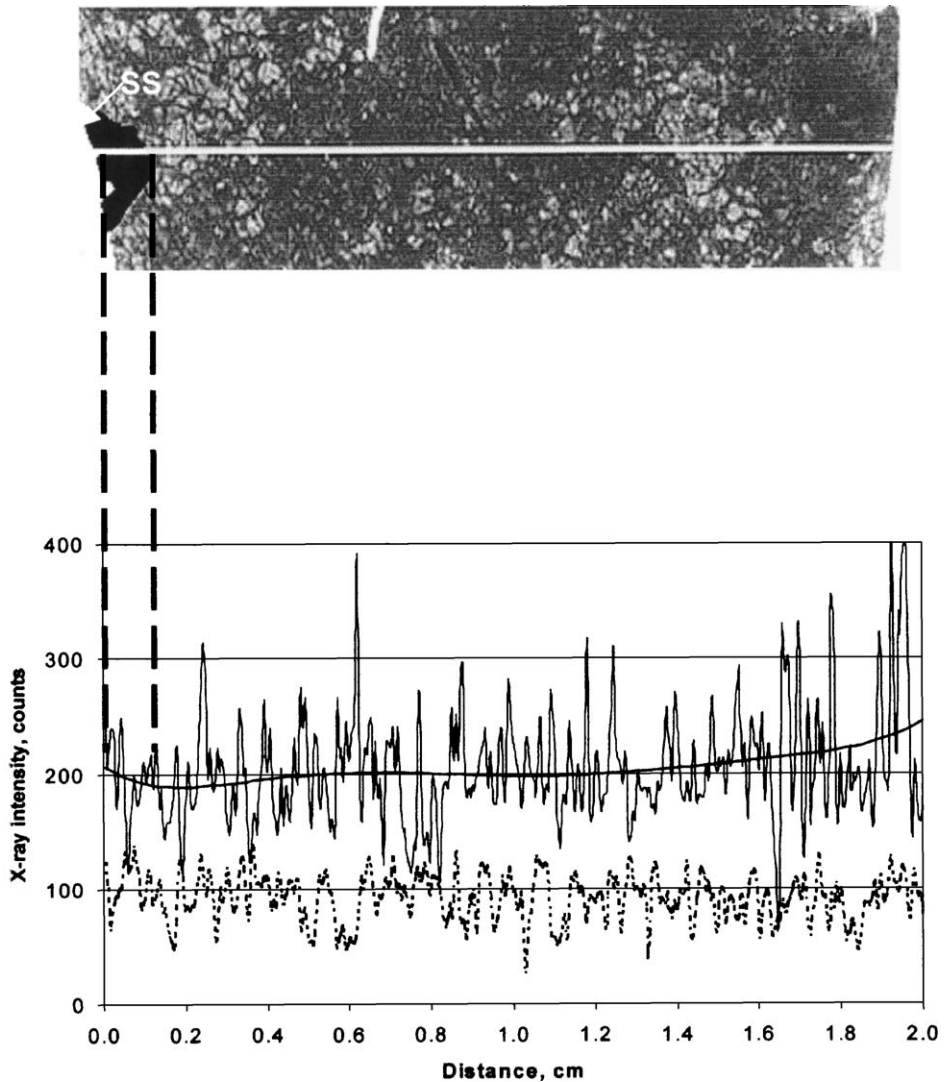


Fig. 12. Illustration of distribution of Brilliant Blue FCF dye (indicated by the bracketed zone) and Br along a linear traverse through a slickenside surface (SS) in the sample from the Bssk horizon (100 cm) of Ships clay. Dotted line indicates antecedent Br concentration.

3.5. Submicroscopic observation of Br distribution through areas adjacent to slickenside planes

Observation of Br distribution in the areas subjacent to the slickenside surfaces in five samples of Ships clay and Burleson clay revealed that the concentration of Br remained essentially constant throughout the sample. Contrary to expectations, concentration of Br was not higher in close vicinity of slickenside plane. In fact, it was often slightly lower in the areas immediately adjacent to the slickenside surface, compared to Br distribution in the remaining portion of the sample (Fig. 12). On the other hand, Brilliant Blue FCF dye penetrated through the surfaces of slickenside fissures only to the extent of 1 to 1.5 mm (Fig. 12).

These observations gave us a reason to suggest that the surfaces of slickenside planes did not serve as significant barriers for Br penetration into the aggregate interior, in spite of unique parallel and subparallel reorientation of clay particles at the slickenside surface. Perhaps the transport of water and solutes across these features was conducted through the cracks and pores that were observed on slickenside surfaces.

4. Conclusions

Based on the dye transport pathways observed, slickenside planes appeared to play a less important role in bypass solute transport in these Vertisols than structural interfaces and biopores. This is contrary to the commonly held view of slickensides serving as an important pathway in bypass solute and water transport. It is possible that the involvement of slickenside planes in solute could increase with the increase in number of open desiccation cracks in Ships and Burleson clay. Likewise, the increase in the volume of applied solutes during seasonal wet periods and high intensity rainfall events could influence solute transport processes in these systems.

It was found that slickenside surfaces differed sufficiently from the underlying matrix because of the stronger preferred orientation of clay particles compared to micromass of the ped interior. In addition, oriented clay particles at the surface of slickenside were characterized by continuity over distances of at least 500 μm . The slickenside surfaces were penetrated by biopores and intersected by coarse cracks.

Microprobe analysis for Br intensity along stained slickenside planes in Burleson clay and Ships clay showed that Br penetrated the surface of these features and became distributed throughout the soil matrix. Noticeable differences were observed between distribution of Brilliant Blue FCF (large organic anion) and Br (smaller inorganic anion). It was found that the movement of the visible dye into the ped interior was significantly retarded in comparison with the distribution of Br. These results suggest that the surfaces of slickensides in these Vertisols did not act as major barriers to Br and water penetration into the soil matrix, in spite of unique parallel and subparallel reorientation of clay particles in the immediate vicinity of the slickenside surface. It is possible that the cracks and pores observed on slickenside surfaces were responsible for the transport of water and solutes across these features.

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